

1. **If Civil Aviation Authority of Nepal (CAAN) appointed you as pavement engineer for the review of design report of Runway of XYZ International Airport submitted by international consultant which is going construction immediately after finalization of design. CAAN instructed you to check the runway which should be operated by B747 aircraft or equivalent (ICAO Code 4E). How can you start your assigned works and what types of elements are you considered? Describe briefly.**

Answer:

For reviewing the runway pavement design of an ICAO Code 4E airport intended to accommodate B747 or equivalent aircraft, I would proceed systematically as follows.

i. **Review of the design basis**

First, I would obtain and thoroughly review the consultant's pavement design report, drawings, specifications, geotechnical report, aircraft forecast and operational requirements.

The first things to verify are:

- a. Design aircraft: B747 or equivalent
- b. ICAO aerodrome reference code: 4E
- c. Design life of pavement
- d. Expected annual aircraft movements
- e. Aircraft load and wheel configuration
- f. Applicable design standards and methodology

The design should be checked against the applicable ICAO standards/recommendations and CAAN requirements, together with the pavement design method adopted by the consultant.

ii. **Check aircraft characteristics and traffic**

Since B747 is the critical aircraft, I would verify:

- Maximum take-off weight (MTOW)
- Maximum landing weight
- Aircraft wheel-load distribution
- Wheel spacing and tyre pressure
- Annual departures/operations
- Traffic growth during the design period

iii. **Review site investigation and subgrade**

The pavement design is only reliable if the subgrade investigation is adequate. Therefore, I would check:

- Soil classification
- CBR or resilient modulus
- Bearing capacity
- Compaction characteristics
- Groundwater level
- Swelling/shrinkage potential

The investigation should adequately represent the entire runway alignment, particularly areas of weak or variable subgrade.

iv. **Check pavement layer design**

For a flexible runway pavement, I would review:

- a. Asphalt/concrete surface course
- b. Binder/base course
- c. Granular or stabilized base
- d. Sub-base
- e. Compacted subgrade

For a **rigid pavement**, I would check:

- a. Concrete slab
- b. Base/sub-base
- c. Subgrade
- d. Joints and joint load-transfer system.

v. **Check pavement structural capacity**

The proposed pavement should be checked for the major aircraft-induced failure modes.

For **flexible pavement**, these include:

- Fatigue cracking
- Rutting
- Permanent deformation
- Shear failure
- Thermal cracking

For **rigid pavement**, important considerations include:

- Slab cracking
- Fatigue
- Faulting
- Pumping
- Joint deterioration

I would independently verify the consultant's calculations rather than accepting only the final pavement thickness.

vi. **Check runway geometric requirements**

Because this is a **Code 4E runway**, I would also review the runway geometry and associated pavement areas, including:

- Runway length
- Runway width
- Shoulder width
- Runway longitudinal and transverse slopes

The pavement should also provide adequate strength for taxiways, turning areas and other aircraft movement areas associated with the runway.

vii. **Examine drainage and environmental conditions**

Adequate drainage is extremely important because water can significantly reduce subgrade strength and accelerate pavement deterioration.

I would check:

- Surface drainage
- Subsurface drainage
- Groundwater control
- Flooding potential
- Rainfall intensity

For Nepal, particular attention should be given to monsoon rainfall, flooding and groundwater conditions.

viii. **Review pavement–aircraft operational compatibility**

I would verify that the pavement can safely support the intended aircraft operations throughout its design life. This includes checking:

- Aircraft wheel loads
- Tyre pressures
- Runway occupancy and traffic distribution
- Heavy aircraft maneuvering
- Possible future aircraft heavier than the present design aircraft

ix. **Check construction and quality-control requirements**

Since construction will start immediately after finalization, I would ensure that the design is practically constructible. I would review:

- Availability of aggregates and construction materials
- Asphalt/concrete mix design
- Plant and equipment requirements
- Compaction requirements
- Quality-control tests

x. **Final independent review**

Finally, I would prepare an independent pavement design review, comparing the consultant's assumptions, calculations and proposed pavement thickness with the applicable requirements.

2. **Make a survey team for detailed survey of the STOL airfield proposed in remote area of the country and make a list that is necessary to carry out the said survey. Also explain the problems that may encounter in carrying out the detailed survey in remote areas.**

Answer:

A STOL (Short Take-Off and Landing) airfield requires a detailed survey to establish the terrain, ground conditions, runway alignment, drainage, obstacles, and other features required for design and construction. For a remote-area airfield, the survey team should be multidisciplinary.



- i. Proposed Survey Team:
A suitable survey team may consist of:

S.N.	Team member	Main responsibility
1	Survey Engineer / Team Leader	Overall planning, coordination and supervision of the survey
2	Senior Surveyor	Control survey, traverse and detailed topographic survey
3	Assistant Surveyors	Levelling, ranging, setting out and field measurements
4	GIS/Remote Sensing Specialist	Preparation of maps, satellite/drone data and GIS analysis
5	Geotechnical Engineer/Engineering Geologist	Soil investigation, rock identification and assessment of ground conditions
6	Hydrologist/Drainage Engineer	Study of drainage pattern, streams, flood levels and catchments
7	CAD/GIS Technician	Processing survey data and preparation of drawings
8	Survey Assistants/Chainmen	Instruments, ranging, marking points and field support
9	Local Liaison/Community Representative	Assistance with local access, land boundaries and communication
10	Field support staff	Transport, camp, equipment, safety and logistics

The lists of other things that are essential for carrying out survey are:

- i. Surveying instruments like GNSS, total stations, prisms, measuring tapes etc.
- ii. Other field materials required such as survey pegs and drawing sheets.

The list of survey to be carried out:

- i. Reconnaissance survey:
- ii. Establishment of horizontal and vertical control
- iii. Detailed topographical survey:
- iv. Longitudinal and cross-sectional survey
- v. Drainage and hydrological survey
- vi. Geological and geotechnical survey

Problems Encountered During Detailed Survey in Remote Areas

- i. Lack of accessibility in remote areas due to steep and rugged topography, rivers, dense forests etc.

- ii. Poor communication facilities making communication with main office and emergency conditions difficult.
- iii. Poor visibility and unfavorable weather
- iv. Difficulties in establishing control points especially in steep, unstable and landslide-prone terrain.
- v. Obstruction on the ground
- vi. Landslides, unstable ground and natural calamities.
- vii. Limited availability of skilled manpower.

3. **Describe briefly about runway, taxiway and apron with necessary sketch.**

Answer:

Runway:

A runway is a defined rectangular area at an aerodrome prepared for the landing and take-off of aircraft. It is generally provided with a strong, smooth and properly drained pavement capable of carrying the loads imposed by aircraft. The runway includes the paved surface and associated shoulders/strip as required.

Main features:

- Used primarily for aircraft take-off and landing.
- Designed according to the type and size of aircraft using the airport.
- Has specified length, width, longitudinal and transverse slopes.
- Requires adequate pavement strength, surface friction and drainage.
- Runway markings, lighting and approach aids are provided as required.

Taxiway:

A taxiway is a defined path provided on an aerodrome for the taxiing of aircraft and for connecting different parts of the airport, particularly the runway with the apron. Taxiways allow aircraft to move safely without using the runway unnecessarily.

Main features:

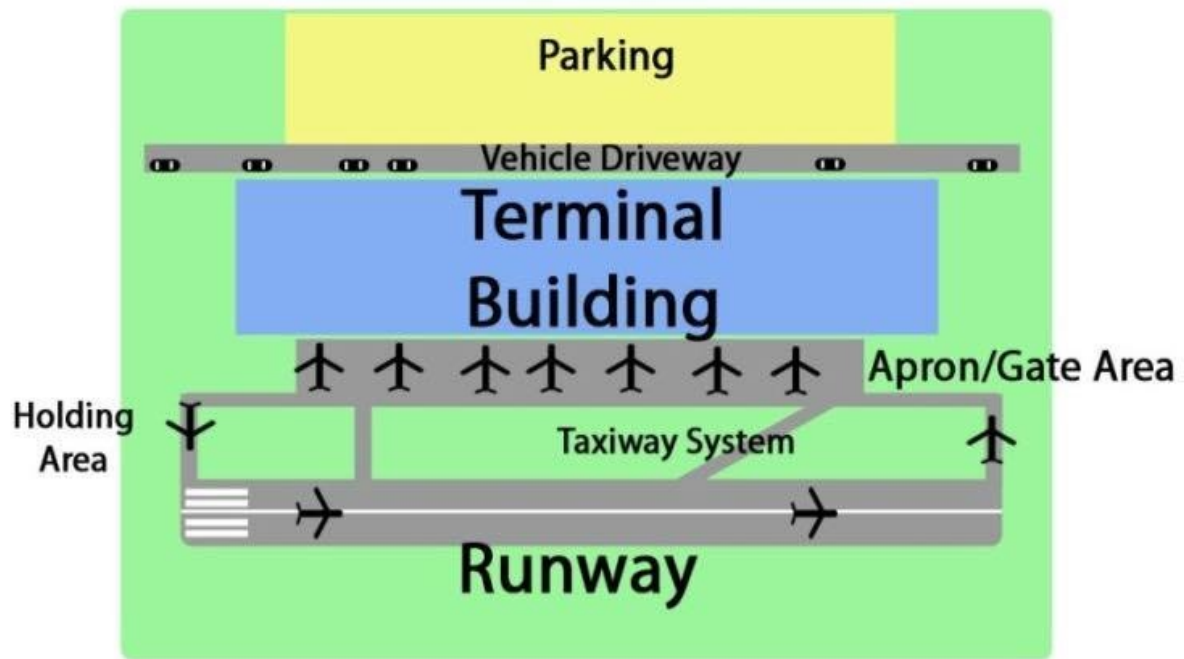
- Connects runway, apron and other airport facilities.
- Provides safe movement of aircraft on the ground.
- Generally, has lower design requirements than the runway, although it must support the aircraft expected to use it.
- Designed considering aircraft turning radius, wheel loads and clearance requirements.
- Proper markings, lighting and signs are provided.

Apron:

An apron is a defined area at an aerodrome intended to accommodate aircraft for loading and unloading passengers or cargo, refueling, parking, maintenance and other ground operations. It is generally located near the terminal building and is connected to the taxiway system.

Main features:

- Used for aircraft parking and ground servicing.
- Provides space for passenger and cargo handling.
- Accommodates aircraft during loading/unloading and refueling.
- Requires sufficient pavement strength to withstand aircraft loads.
- Must provide adequate space for aircraft maneuvering and clearance.



4. **Describe the basic steps to be considered for the airport site selection.**

Answer:

Airport site selection is the process of identifying and evaluating a suitable location for constructing an airport so that it is safe, technically feasible, economically viable and environmentally acceptable.

The basic steps are:

i. **Define the airport requirements:**

First, determine:

- Type of airport and its intended function
- Required runway length and orientation
- Expected passenger and cargo traffic
- Future expansion requirements

ii. **Preliminary study and reconnaissance**

Collect existing information and conduct a preliminary study of the region, including:

- Topographic maps
- Geological and soil maps
- Land-use maps
- Meteorological data
- Population and settlement information

Potential sites are then identified for further investigation.

iii. **Study of topography and terrain**

The proposed site should be examined for:

- Ground elevation and slope
- Hills, valleys and steep terrain
- Natural drainage
- Earthwork requirements
- Possibility of runway development and expansion

iv. **Meteorological investigation**

Long-term weather data should be studied, particularly:

- Wind direction and velocity
- Wind rose
- Rainfall
- Fog and visibility
- Cloud conditions
- Temperature

v. **Obstacle and airspace study**

Identify natural and artificial obstacles such as:

- Hills and mountains
- Trees
- Buildings
- Towers and communication masts
- Transmission lines

vi. **Soil and geological investigation**

The site should be investigated for:

- Soil type and bearing capacity
- Subgrade strength
- Groundwater level
- Landslide susceptibility
- Settlement potential

The site should have suitable ground conditions for airport pavements and structures.

vii. **Accessibility and transportation**

The airport should have good connectivity with the service area through:

- Highways/roads
- Public transportation
- Railways, where available

viii. **Land availability and future expansion**

Consider:

- Total land requirement
- Land ownership
- Cost of land acquisition
- Existing settlements and structures
- Possibility of future runway and terminal expansion

Adequate land should be available for both present requirements and future development.

ix. **Environmental and social assessment**

Potential impacts on:

- Forests
- Wildlife
- Water bodies

- Noise-sensitive areas
- Cultural and archaeological sites should be evaluated.

x. **Economic and financial evaluation**

Estimate:

- Land acquisition cost
- Construction cost
- Earthwork cost
- Drainage requirements
- Operating and maintenance costs

Alternative sites should be compared based on their overall economic feasibility.

xi. **Safety and security assessment**

The site should be evaluated for:

- Aircraft operational safety
- Bird-strike hazards
- Flooding
- Landslides
- Strong winds and other natural hazards

xii. **Detailed survey and comparison of alternatives**

After preliminary screening, detailed surveys and investigations should be carried out for the most promising sites. A comparative analysis is then prepared based on technical, operational, environmental, social and economic factors.

xiii. **Final site selection**

Finally, the most suitable site is selected and recommended for approval after considering all major factors and applicable ICAO and national aviation requirements.

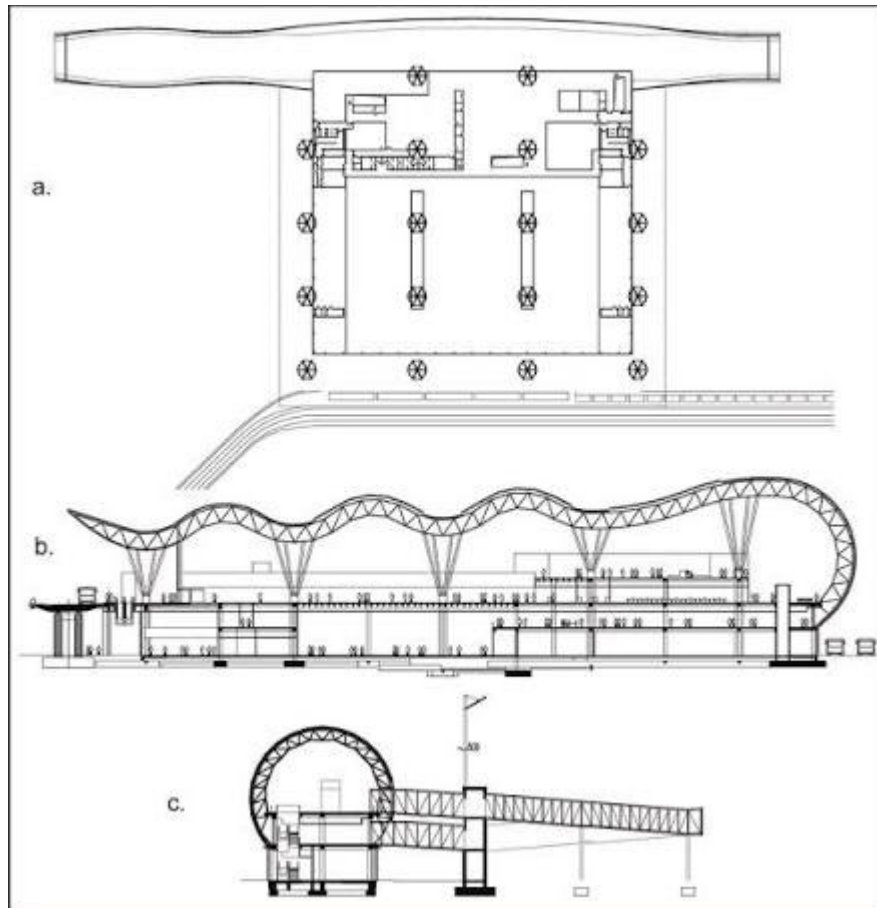
5. **What is project management? Describe the various phases of the terminal building?**

Answer:

Project management is the systematic process of planning, organizing, coordinating, executing, monitoring and controlling the resources and activities of a project to achieve specified objectives within the required time, cost and quality.

In the construction of an airport terminal building, project management ensures that the project is completed safely, economically, on schedule and according to the required standards.

Phases of a Terminal Building Project:



The major phases involved in the development and construction of an airport terminal building are:

i. **Planning and Feasibility Phase:**

This is the initial phase in which the need and feasibility of the terminal are established.

- Estimation of passenger and aircraft traffic.
- Site selection and site assessment.
- Study of economic and financial feasibility.
- Identification of project objectives and scope.
- Preparation of the initial project schedule.

ii. **Design Phase**

In this phase, the detailed design of the terminal building is prepared.

- Architectural planning and layout.
- Structural design.
- Electrical and mechanical design.
- Passenger circulation and baggage-handling systems.
- Preparation of drawings, specifications and cost estimates.

The design should comply with applicable **aviation, building, safety and accessibility standards**.

iii. **Procurement and Tendering Phase**

After completion of the design, contractors and suppliers are selected.

Major activities are:

- Preparation of tender documents.
- Preparation of bill of quantities.
- Invitation of bids.
- Evaluation of bids.

iv. **Construction Phase**

Major works include:

- Site clearance and preparation.
- Earthwork and foundation construction.
- Roofing and building envelope.
- Internal and external finishes.
- Installation of electrical and mechanical systems.

Project management during this phase involves controlling time, cost, quality, safety and resources.

v. **Testing and Commissioning Phase**

After construction, all systems and facilities are tested to ensure proper operation.

This includes:

- Electrical system testing.
- HVAC testing.
- Water supply and drainage testing.
- Baggage-handling system testing.
- Security system testing.

Defects identified during testing are corrected before opening.

vi. **Handover and Opening Phase**

After successful commissioning:

- The completed facility is inspected.
- As-built drawings and operation manuals are handed over.
- Necessary approvals and certificates are obtained.
- Airport staff are trained.
- The terminal is handed over to the operator.

vii. **Operation and Maintenance Phase**

After opening, the terminal enters its operational phase.

- Routine operation and maintenance.
- Inspection of building and equipment.
- Repair and replacement of defective components.
- Maintenance of electrical, mechanical and safety systems.
- Monitoring passenger facilities.
- Periodic upgrading and expansion according to future demand.

6. **What are the facilities that may be required for the planning and design of the terminal building?**

Answer:

While planning and designing an airport terminal building, facilities should be provided to ensure the safe, efficient and comfortable movement of passengers, baggage, aircraft and airport staff. The required facilities may include:



i. **Passenger Facilities**

- Arrival and departure halls
- Check-in counters
- Waiting and seating areas
- Ticketing facilities
- Immigration facilities for international airports
- Customs facilities
- Security screening areas
- Boarding gates
- Departure lounges
- Arrival lounges
- Toilets and washrooms
- Facilities for persons with disabilities

ii. **Baggage Facilities**

- Baggage check-in and conveyor systems
- Baggage handling area
- Baggage storage
- Baggage inspection and security screening

iii. **Aircraft-Related Facilities**

- Aircraft parking stands
- Apron area
- Passenger boarding bridges, where required
- Ground-support equipment areas
- Aircraft servicing facilities
- Refueling facilities
- Catering service areas

iv. **Airport Security Facilities**

- Passenger and baggage security screening
- Security control rooms
- CCTV monitoring
- Access-control systems
- Security offices
- Emergency exits and evacuation facilities
- Fire detection and fire-fighting systems

v. **Commercial Facilities**

- Restaurants and cafés
- Shops
- Duty-free shops
- Banks and ATMs
- Business/VIP lounges

vi. **Administrative Facilities**

- Airport administration offices
- Airline offices
- Staff rooms
- Operations control room
- Security administration
- Staff toilets and changing rooms

vii. **Transportation and Access Facilities**

- Approach/access roads
- Passenger drop-off and pick-up areas
- Car parking
- Taxi and bus parking
- Public transportation facilities
- Service and emergency vehicle access

viii. **Utility and Service Facilities**

- Electrical substation and backup power
- Water supply system
- Water storage tanks
- Sewage and wastewater facilities
- Solid-waste management facilities
- HVAC system

7. **What is pavement evaluation and why is it necessary? Explain briefly the pavement evaluation process. Describe the meaning of PCN 80/R/V/N/T.**

Answer:

Pavement evaluation is the systematic process of determining the structural condition, functional condition and load-carrying capacity of an existing airport pavement and assessing whether it is suitable for continued aircraft operations.

Pavement evaluation is necessary to:

- Determine the present condition and strength of the pavement.
- Check whether the pavement can safely carry the existing and future aircraft traffic.
- Identify defects such as cracking, rutting, deformation, surface deterioration and unevenness.
- Determine the need for maintenance, rehabilitation or strengthening.
- Prevent premature pavement failure and reduce maintenance costs.
- Ensure safe aircraft operation.
- Determine and report the pavement's PCN (Pavement Classification Number).

The pavement evaluation process can be broadly carried out in the following steps:

i. **Collection of Existing Information**

Collect available information regarding:

- Pavement type and layer thicknesses
 - Original pavement design
 - Construction history
 - Previous maintenance and rehabilitation
 - Aircraft traffic and loading.
- ii. Visual Inspection:
Conduction of a detailed field inspection to identify and record cracks, potholes, ruts, surface deformation, joint deterioration etc.
- iii. Functional Evaluation:
The pavement surface is assessed for characteristics affecting aircraft operation, such as surface roughness, skid resistance, drainage, surface texture etc.
- iv. Structural Evaluation

Determination of the structural capacity of the pavement using methods such as plate load test, deflection test, Falling Weight Deflectometer etc.
- v. Material and Subgrade Investigation

Where necessary, collect samples or conduct tests to determine:
- Asphalt/concrete properties
 - Base and sub-base characteristics
 - Subgrade CBR or other strength parameters
 - Layer thickness
 - Moisture and drainage conditions
- vi. Analysis and Assessment

The collected data are analyzed to determine the pavement conditions, suitability for existing and proposed aircraft, critical areas and failure mechanisms.
- vii. Maintenance/Rehabilitation Recommendation

Finally, appropriate and required measures are recommended, such as routine maintenance, repair, overlay, rehabilitation etc.

PCN 80/R/V/N/T is an example of an ICAO Pavement Classification Number reported using the ACN-PCN system.

It means that the pavement has a PCN of 80, is a rigid pavement, has a high-strength subgrade, has no tyre-pressure limitation, and its strength has been determined by technical evaluation.

Notes:

Codes used in PCN Reporting:

Pavement type:

R = Rigid

F = Flexible

Subgrade category:

A = High strength

B = Medium strength

C = Low strength

D = Ultra-low strength

Tyre-pressure category:

W = No limit

X = Up to 1.50 MPa

Y = Up to 1.00 MPa

Z = Up to 0.50 MPa

Evaluation method:

T = Technical evaluation

U = Using aircraft experience